

ANIKA TABASSUM SEJUTY

Seodaemun-gu, Seoul, South Korea | +821042533876 | sejuty1111@gmail.com
LinkedIn | Google Scholar

PROFILE

AI researcher with expertise spanning deep learning, computer vision, medical imaging, and signal processing. Experienced in developing end-to-end learning-based solutions for ultrasound imaging, from data representation and algorithm development to experimental validation using simulated and clinical datasets. Published peer-reviewed research and presented at international conferences, with a strong interest in translating advanced AI research into reliable healthcare technologies.

EDUCATION

Integrated Master's-Doctoral in Electronics & Electrical Engineering *[Fall 2023 – Present]*

Ewha Womans University, Seoul, Republic of Korea

GPA: 4.12/4.3

Bachelor of Science in Computer Science & Engineering *[2022]*

Northern University of Business & Technology Khulna, Bangladesh

GPA: 3.898/4.0

Undergraduate Thesis:

Machine Learning-Based Alzheimer's Disease Prediction from Human Behavior

- Developed and compared supervised machine learning models including SVM, KNN, Random Forest, XGBoost, AdaBoost, and Voting Classifier
- Evaluated model performance for Alzheimer's disease prediction using behavioral data.

RESEARCH EXPERIENCE

Graduate Research Assistant *[September 2023 – Present]*

Intelligent Sensing & Processing Lab | Ewha Womans University

- Developed deep learning-based motion tracking methods for subpixel estimation of vascular wall motion from ultrasound imaging.
- Implemented and evaluated multiple Siamese neural network architectures using simulated and in vivo carotid ultrasound datasets.
- Conducted comparative evaluation of RF, I/Q (modulated and demodulated), and B-mode ultrasound data representations to assess their impact on deep learning-based motion tracking performance.
- Presented research at six international and domestic conferences and published in peer-reviewed venues including Biomedical Signal Processing and Control and IEEE Robotics and Automation Letters.

Research Assistant *[June 2022 – July 2023]*

Department of Research & Development | Pioneer Alpha Ltd. (Remote)

- Improved CNN model performance through architecture design and optimization.
- Conducted literature review and thematic analysis of recommender systems research.
- Identified research gaps and proposed future research directions for AI applications in COVID-19.

PUBLICATIONS

C-Siam: A Complex-Valued Siamese Network for Ultrasound-Based Vascular Motion Estimation
Biomedical Signal Processing and Control, 2026

Authors: **A. T. Sejuty**, N. C. P. Nguyen, S. Park

Sensorless Force Imbalance Prediction Based on Visual Information for Robotic Ultrasound Scanning
IEEE Robotics and Automation Letters, 2025

Authors: Q. Liu, E. Choi, **A. T. Sejuty**, S. Park

2D Motion Tracking for Vascular Wall in Ultrasound Imaging

Proceedings of International Conference on Electronics, Information, and Communication, 2024

Authors: **A. T. Sejuty**, J. Seo, S. Park

Generative Adversarial Networks for Data Augmentation

Book Chapter in Data Driven Approaches on Medical Imaging, 2023

Authors: A. Biswas, N. M. Abdullah Al, A. Imran, **A. T. Sejuty**, F. Fairouz, S. Puppala, S. Talukder

CONFERENCE PRESENTATIONS

Extended Analysis of Siamese Networks for Complex Ultrasound Data

66th Korean Society of Medical & Biomedical Engineering Fall Academic Conference (Poster), 2025

Deep Complex Network for Vascular Wall Motion Tracking in Ultrasound Imaging

IEEE International Ultrasonics Symposium (IUS) (Poster), 2025

Assessing Complex Data Representations in Siamese Networks for Vascular Wall Motion Tracking

International Conference on Ultrasound Engineering for Biomedical Applications (ICUEBA) (Poster), 2025

Comparative Analysis of Ultrasound Data Representations for Vascular Wall Motion Tracking

65th Korean Society of Medical & Biomedical Engineering Spring Academic Conference (Poster), 2025

Tracking 2D Vascular Wall Motion in In Vivo Ultrasound Imaging

IEEE Ultrasonics, Ferroelectrics, and Frequency Control Joint Symposium (Poster), 2024

2D Motion Tracking for Vascular Wall in Ultrasound Imaging

International Conference on Electronics, Information, and Communication (Oral), 2024

TECHNICAL SKILLS

Programming & Tools:

Python, MATLAB, Git

Deep Learning & Computer Vision:

PyTorch, OpenCV, CNNs, Siamese Networks, U-Net, Complex-Valued Neural Networks, Image Segmentation, Motion Tracking

Medical Imaging & Signal Processing:

Ultrasound Imaging, RF & I/Q Signal Processing, Delay-and-Sum (DAS) Beamforming, Feature Extraction, Image & Video Coding

Research Platforms:

Verasonics Vantage-256, Row-Column Array Ultrasound Imaging

HONORS & SCHOLARSHIPS

- International Students Scholarship (TQ3), awarded throughout graduate coursework, 2023 – 2026
- Research Assistant Scholarship, 2025 – Present
- Brain Korea 21 (BK21) Scholarship, 2024
- Student Assistant Scholarship, 2024
- University Merit Scholarship, awarded across multiple undergraduate semesters